

5.2 Human Detector Playground

The demo further offers a human detector playground as an interactive interface, which allows users to test their capability to distinguish between the four text categories. Figure 1 (bottom) shows a snapshot of the playground interface where the users can try the system, gaining insights into the subtle differences between various types of human-written and machine-generated texts.

5.3 Deployment and Implementation

Our demo is deployed on Hugging Face Spaces, which allows seamless integration with transformer models, ease of use, and robust support for hosting machine learning applications. For implementing the user interface, we used Gradio. The code is publicly available under an MIT license.

6 Conclusion and Future Work

In an era of advanced large language models, maintaining the integrity of text poses significant challenges. We presented a system that aims to identify the use of machine-generated text, accurately differentiating human-written text from various types of automatically generated text. Unlike previous work, we use a fine-grained classification schema (Human-Written, Machine-Generated, Machine-Written Machine-Humanized, and Human-Written Machine-Polished), which offers insights into the origins of the text, thus enabling trustworthiness.

In future work, we plan to improve the Domain Adversarial Neural Network (DANN) to improve the results even further. We further plan to explore the possibility of using a DANN on the text’s generator instead of the text’s domain to generalize detection across different text generators. Using a DANN on both the domain and the generator could potentially lead to a truly universal detector. We also aim to expand the classification to include a fifth category: machine-written and human-edited text, enhancing detection capabilities and providing a more comprehensive analysis of text origins. To further improve the system, we also plan to address potential biases in the dataset caused by formatting styles linked to specific domains, such as Wikihow and PeerRead, to ensure better robustness across a broader range of human-written content. Last but not least, we want to expand the dataset to encompass a diverse set of languages, enabling the development of a robust multilingual detection model.

Limitations

We acknowledge certain limitations of our work, which we plan to address in future work. First, although our work has explored more fine-grained machine-generated text scenarios beyond conventional binary classification, we did not consider a complex scenario where the text is first generated by a machine and then is manually edited by humans to suit their personal needs. This is primarily due to the high costs associated with collecting data that requires human editing.

Moreover, we identified some issues with the dataset. Specifically, some LLMs associate specific domains with particular formatting styles, such as markdown for lists, bullet points, and headers. This issue was particularly noticeable in the Wikihow and PeerRead domains, where the LLMs frequently applied these formatting styles, potentially skewing the data and impacting the accuracy of our classifications. It also remains uncertain whether our system can generalize to detecting models or languages not included in our English-only dataset.

Ethical Statement and Broad Impact

Data License A primary ethical consideration is the data license. We reused pre-existing corpora, such as OUTFOX and Wikipedia, which have been publicly released and approved for research purposes. Moreover, we generated new data on top of the original data, thereby mitigating concerns regarding data licensing.

Biased and Offensive Language Considering that our data is generated by large language models, it might contain offensive or biased language; we did not try to control for this, relying on the internal safety mechanisms of the LLMs we used.

Positive Impact of Fine-Grained Detection

LLM-DetectAIve expands the conventional binary classification in machine-generated text detection to more fine-grained levels, which is more aligned with real-life scenarios. We believe this approach could be applied in various scenarios, e.g., for students’ essays to ensure the originality of their work. Moreover, LLM usage detection may find applications in authorship detection as well as in digital forensics.

References

- Anthropic. 2024. [The Claude 3 model family: Opus, Sonnet, Haiku](#).
- Evan N Crothers, Nathalie Japkowicz, and Herna L Viktor. 2023. Machine-generated text: A comprehensive survey of threat models and detection methods. *IEEE Access*, 11:70977–71002.
- Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, and Victor S. Lempitsky. 2017. [Domain-adversarial training of neural networks](#). In Gabriela Csurka, editor, *Domain Adaptation in Computer Vision Applications*, Advances in Computer Vision and Pattern Recognition, pages 189–209. Springer.
- Team Gemini. 2023. [Gemini: A family of highly capable multimodal models](#). *arXiv:2312.11805*.
- Pengcheng He, Xiaodong Liu, Jianfeng Gao, and Weizhu Chen. 2021. [DeBERTa: Decoding-enhanced BERT with disentangled attention](#). In *Proceedings of the International Conference on Learning Representations*.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. [Mistral 7b](#). *arXiv:2310.06825*.
- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, et al., Szymon Antoniak, Teven Le Scao, Th  ophile Gervet, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2024. [Mixtral of experts](#). *arXiv:2401.04088*.
- Ryuto Koike, Masahiro Kaneko, and Naoaki Okazaki. 2024. [OUTFOX: LLM-generated essay detection through in-context learning with adversarially generated examples](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 21258–21266.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. [RoBERTa: A robustly optimized BERT pretraining approach](#). *arXiv:1907.11692*.
- Team Llama. 2024. [The Llama 3 herd of models](#). *arXiv:2407.21783*.
- Dominik Macko, Robert Moro, Adaku Uchendu, Jason Lucas, Michiharu Yamashita, Mat   s Pikuliak, Ivan Srba, Thai Le, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2023. [MULTITuDE: Large-scale multilingual machine-generated text detection benchmark](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 9960–9987, Singapore.
- Dominik Macko, Robert Moro, Adaku Uchendu, Ivan Srba, Jason Samuel Lucas, Michiharu Yamashita, Nafis Irtiza Tripto, Dongwon Lee, Jakub Simko, and Maria Bielikova. 2024. [Authorship obfuscation in multilingual machine-generated text detection](#). *arXiv:2401.07867*.
- Eric Mitchell, Yoonho Lee, Alexander Khazatsky, Christopher D Manning, and Chelsea Finn. 2023. [DetectGPT: Zero-shot machine-generated text detection using probability curvature](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202, pages 24950–24962. PMLR.
- OpenAI. 2023. [GPT-4 technical report](#). *arXiv:2303.08774*.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. [DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter](#). *ArXiv:1910.01108*.
- Teo Susnjak. 2022. [ChatGPT: The end of online exam integrity?](#) *arXiv:2212.09292*.
- Ruixiang Tang, Yu-Neng Chuang, and Xia Hu. 2024. [The science of detecting LLM-generated text](#). *Communications of the ACM*, 67(4):50–59.
- Gemma Team. 2024. [Gemma 2: Improving open language models at a practical size](#). *arXiv:2408.00118*.
- Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Osama Mohammed Afzal, Tarek Mahmoud, Giovanni Puccetti, and Thomas Arnold. 2024a. [SemEval-2024 task 8: Multidomain, multimodel and multilingual machine-generated text detection](#). In *Proceedings of the 18th International Workshop on Semantic Evaluation*, pages 2057–2079, Mexico City, Mexico.
- Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Osama Mohammed Afzal, Tarek Mahmoud, Giovanni Puccetti, Thomas Arnold, Alham Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2024b. [M4GT-bench: Evaluation benchmark for black-box machine-generated text detection](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, pages 3964–3992, Bangkok, Thailand.
- Yuxia Wang, Jonibek Mansurov, Petar Ivanov, Jinyan Su, Artem Shelmanov, Akim Tsvigun, Chenxi Whitehouse, Osama Mohammed Afzal, Tarek Mahmoud, Toru Sasaki, Thomas Arnold, Alham Fikri Aji, Nizar Habash, Iryna Gurevych, and Preslav Nakov. 2024c. [M4: Multi-generator, multi-domain, and multi-lingual black-box machine-generated text detection](#). In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics*, pages 1369–1407, St. Julian’s, Malta.
- Qihui Zhang, Chujie Gao, Dongping Chen, Yue Huang, Yixin Huang, Zhenyang Sun, Shilin Zhang, Weiye Li, Zhengyan Fu, Yao Wan, and Lichao Sun. 2024. [LLM-as-a-coauthor: Can mixed human-written and machine-generated text be detected?](#) In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 409–436, Mexico City, Mexico. Association for Computational Linguistics.